An Internship Report

Of

Data Science Master Virtual Internship

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech (CSE)



By

Roll Number of Student: 2410030778

Name of Student: KARTIKEYA KHARE

Batch: 3CSE12 (2024-2028)

Batch:

2026-2027

CANDIDATE'S DECLARATION

I, Kartikeya Khare, hereby declare that the internship report titled “Siemens Data Science Master Virtual Internship” is an original record of the work carried out by me and submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering.

This report documents the learning, activities, and outcomes of the 8-Week AICTE–EduSkills Virtual Internship Program undertaken in the domain of Siemens Data Science Master, carried out between April 2026 and June 2026.

All sources of information used while preparing this report have been duly acknowledged. I further declare that this work has not been submitted, in part or in full, to any other institute or university for the award of any other degree or diploma.

I take full responsibility for the authenticity and accuracy of the information presented in this report.

Signature:

Student Name: Kartikeya Khare
Roll No.: 2410030778

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who guided and supported me throughout the course of this internship. Their encouragement and constructive feedback played an important role in helping me complete the program and deepen my understanding of Data Science and its allied technologies.

I am grateful to the AICTE–EduSkills Virtual Internship Program team and EduSkills Academy for offering me the opportunity to be part of the Siemens Data Science Master Virtual Internship. The eight-week, week-wise learning structure gave me a well-organised path to build industry-relevant knowledge in data engineering, machine learning, data preparation, applications and use cases, and platform administration.

My sincere thanks also go to IILM University, Greater Noida, and to the faculty of the School of Computer Science and Engineering for their continued guidance and encouragement throughout this internship.

The structured weekly modules, assessments, and the final evaluation helped me stay consistent in my learning and gradually connect Data Science and Machine Learning theory with practical, industry-oriented use cases.

Finally, I am thankful to my family and friends for their constant support and motivation throughout this journey.

Signature:

Student Name: Kartikeya Khare
Roll No.: 2410030778

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3. INTERNSHIP COMPLETION CERTIFICATE

The Internship Offer Letter issued by EduSkills Academy confirms the selection of Kartikeya Khare, B.Tech (Computer Science and Engineering), IILM University, Greater Noida, for the 8-Week AICTE–EduSkills Virtual Internship Program in the Siemens Data Science Master domain for the period April 2026 to June 2026.

As specified in the offer letter, successful completion of the program required following a week-wise structured learning plan, clearing weekly assessments, submitting project documentation and assignments where applicable, and appearing for the Final Assessment Test.

The Certificate of Virtual Internship and the corresponding Internship Offer Letter, both issued in my name, are reproduced below as supporting documents.

Internship Offer Letter

The offer letter (Ref No. C16Y2026M091397690) confirms my selection for the 8-week Siemens Data Science Master Virtual Internship, conducted from April 2026 to June 2026, and outlines the week-wise module structure, the weekly assessments, and the final evaluation required for successful completion.

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Ref No: C16Y2026M091397690

Internship Offer Letter

Student Name	: Kartikeya Khare
Stream	: B.Tech
Branch	: B.Tech(Computer Science and Engineering)
Institute Name	: IILM University, Greater Noida
Internship Domain	: Siemens Data Science Master
Internship Duration	: April 2026 - June 2026
AICTE Student ID	: STU6998573a168e01771591482

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

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The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Applications & Use Cases Professional	Learn to identify business problems suitable for AI and machine learning.
Week 2	Data Engineering Professional	Learn to access, load, transform, and calculate data.
Week 3	Machine Learning Professional	Build foundational machine learning skills.
Week 4	Data Preparation Master	Master advanced data handling and automation.
Week 5	Machine Learning Master	Learn advanced machine learning techniques.
Week 6	Applications & Use Cases Master	Deploy and monitor ML models.
Week 7	Platform Administration Master	Configure and manage RapidMiner AI Hub.
Week 8	Platform Administration Master	Enterprise-level platform operations.
Final Credential Validation / Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.

Bibek Ranjan
Director, EduSkills

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Certificate of Virtual Internship

The certificate below (Certificate ID: 469bb51ab4e288ddd94a, Student ID: STU6998573a168e01771591482) confirms that I successfully completed the 8-week Data Science Master Virtual Internship, supported by Siemens, during April–June 2026, with an Internship Grade of "O" (Outstanding).



4. PROJECT DESCRIPTION

4.1 Introduction

This report describes the 8-week Siemens Data Science Master Virtual Internship that I completed under the AICTE–EduSkills Virtual Internship Program between April 2026 and June 2026, as part of my Bachelor of Technology program at IILM University, Greater Noida.

The internship was designed to build a structured understanding of Data Science, Data Engineering, Machine Learning, Data Preparation, Applications and Use Cases, and Platform Administration through a progressive, week-wise curriculum. It walked me through the different stages of a typical Data Science lifecycle — starting with identifying real-world use cases and moving through data engineering, core and advanced machine learning, model deployment and monitoring, and finally enterprise platform administration.

Learning was reinforced through weekly assessments and, where applicable, project assignments, culminating in a Final Assessment Test. This combination of guided theory and hands-on practice helped me build both technical depth and a practical sense of how Data Science projects are executed in an industry setting.

4.2 Organization Profile

EduSkills Academy runs the AICTE–EduSkills Virtual Internship Program, which gives students access to industry-aligned virtual internship tracks across a range of technology domains, complementing regular academic coursework with practical, hands-on exposure.

I opted for the Siemens Data Science Master track, an eight-week program supported by Siemens, where each week was mapped to a dedicated professional or master-level module on the RapidMiner platform. The program included weekly assessments and concluded with a final credential validation and internship grade assessment, giving a clear and measurable structure to the learning path.

4.3 Internship Problem Statement

Modern organizations generate enormous amounts of data across their operations, and the real challenge lies in converting that raw data into insights that can genuinely support decision-making and power intelligent applications.

As a student, I had studied databases, data processing, machine learning, and artificial intelligence largely as separate subjects. What this internship addressed was the gap between that siloed knowledge and the way these pieces actually fit together in a real, end-to-end Data Science workflow — from spotting a suitable business problem, to sourcing and preparing data, to building and refining models, to eventually deploying and monitoring them on a live platform.

The program was structured around five broad pillars — Applications & Use Cases, Data Engineering, Machine Learning, Data Preparation, and Platform Administration — each covered at both a professional and a master level.

4.4 Internship Objectives

The internship was built around the following objectives:

- Understand the fundamental concepts and real-world applications of Data Science.
- Learn to identify business problems that can be solved using Artificial Intelligence and Machine Learning.
- Understand how data is accessed, loaded, transformed, and calculated.
- Build foundational knowledge of Machine Learning models and workflows.
- Gain proficiency in advanced data preparation and handling techniques.
- Study advanced Machine Learning techniques such as parameter optimization and time-series forecasting.
- Understand how Machine Learning models are deployed for real-world use.
- Learn methods used to monitor deployed Machine Learning models.
- Gain working knowledge of platform configuration and administration on RapidMiner AI Hub.
- Understand the role enterprise platforms play in supporting Data Science and AI workflows.
- Develop industry-relevant technical skills and strengthen analytical and problem-solving ability.
- Successfully complete all weekly assessments and required assignments.
- Clear the Final Credential Validation / Internship Grade Point Assessment.

As stated in the offer letter, the program was designed to combine structured learning with practical exposure, with successful completion of all required assessments leading to the award of the Virtual Internship Certificate.

4.5 Scope of the Internship

The internship spanned the major functional areas of the Data Science ecosystem. Applications and Use Cases covered how Artificial Intelligence and Machine Learning are applied to solve real business problems. Data Engineering dealt with accessing, loading, transforming, and working with raw data. Machine Learning covered both foundational and advanced techniques for learning patterns from data. Data Preparation focused on advanced data handling and automation needed to support reliable analysis. The Applications & Use Cases Master module introduced model deployment and monitoring, while Platform Administration Master covered the configuration and

management of enterprise AI platforms. The entire program was delivered virtually, with an emphasis on structured, assessment-driven learning.

4.6 Technologies and Tools Used

- **Data Science:** An interdisciplinary field combining data processing, statistical analysis, Machine Learning, and domain expertise to extract useful insight from data.
- **Data Engineering:** Concerned with accessing, loading, transforming, and preparing data so it is ready for analysis and modelling.
- **Machine Learning:** The practice of building models that learn patterns from data to perform tasks such as classification, regression, and clustering.
- **Data Preparation:** Techniques for organizing and transforming raw, often messy, data into a clean format suitable for analysis and modelling.
- **Artificial Intelligence Applications:** Applying AI and Machine Learning techniques to practical business and real-world problems.
- **Model Deployment and Monitoring:** Concepts and practices around putting trained models into production and tracking their performance over time.
- **Platform Administration:** Configuration and management of the RapidMiner AI Hub and related enterprise AI infrastructure.
- **Assessment and Learning Platform:** Weekly assessments, assignments, and a final assessment used throughout the program to evaluate progress.

4.7 Data Science Workflow / System Architecture

The internship's learning architecture can be summarised as a pipeline moving from problem identification through to deployment and monitoring:

Business Problem Identification
Data Access and Collection
Data Loading
Data Transformation
Data Preparation
Machine Learning Model Development
Model Evaluation
Model Deployment
Model Monitoring
Application / Business Use Case

This pipeline maps onto seven conceptual layers covered across the program:

- **Problem Identification Layer:** Starting point where a suitable business or real-world problem for AI/ML is identified.

- **Data Engineering Layer:** Relevant data is accessed, loaded, managed, and transformed.
- **Data Preparation Layer:** Raw data is cleaned and shaped for analysis or modelling.
- **Machine Learning Layer:** Prepared data is used to build and refine models.
- **Application Layer:** Trained models are applied to real business use cases.
- **Deployment and Monitoring Layer:** Models are deployed and their performance tracked, as covered in the Applications & Use Cases Master module.
- **Platform Administration Layer:** The AI/Data Science workflow is supported by configuring and managing the underlying enterprise platform.

4.8 Methodology

The internship followed a structured, week-wise methodology, moving from foundational to advanced topics across eight learning weeks, followed by final credential validation and grade assessment. The approach combined:

- Week-wise conceptual learning
- Hands-on practical exposure to each concept
- Weekly assessments to reinforce understanding
- Assignments and project documentation where applicable
- A gradual progression from professional-level to master-level modules
- Final credential validation and Internship Grade Point Assessment

As confirmed in the offer letter, the final grade was based on overall performance across weekly assessments, project submissions, and the final test.

4.9 Week-wise Internship Activities

Week	Module	Key Learning / Activities
Week 1	Applications & Use Cases Professional	Learned to identify business problems suitable for AI and Machine Learning and explored their real-world applications.
Week 2	Data Engineering Professional	Learned to access, load, transform, and calculate data as preparation for analysis and modelling.
Week 3	Machine Learning Professional	Built foundational Machine Learning skills, covering core models, workflows, and evaluation methods.
Week 4	Data Preparation Master	Mastered advanced data handling, transformation, and automation techniques.
Week 5	Machine Learning Master	Studied advanced Machine Learning techniques and how data preparation choices affect model performance.

Week	Module	Key Learning / Activities
Week 6	Applications & Use Cases Master	Learned practical model deployment and monitoring approaches for production use cases.
Week 7	Platform Administration Master	Learned platform configuration and management, with exposure to RapidMiner AI Hub.
Week 8	Platform Administration Master	Studied enterprise-level platform operations, administration, and reliable AI infrastructure.
Final Assessment	Credential Validation / Internship Grade Point Assessment	Completed all weekly assessments, applicable assignments, and the final assessment for internship completion and certification.

4.10 Expected Outcomes and Learning Outcomes

This internship gave me practical grounding across the Data Science lifecycle. I developed a working understanding of Data Science, Data Engineering, Artificial Intelligence, and Machine Learning — covering data access and transformation, data preparation, advanced modelling techniques, and how models are deployed and monitored in production. I also gained exposure to AI platform configuration and enterprise-level administration.

More broadly, the program sharpened my ability to approach problems through a structured pipeline: Problem → Data → Preparation → Model → Application → Deployment → Monitoring. It also strengthened my self-learning discipline, time management, documentation skills, and my overall comfort with industry-style, assessment-driven learning.

4.11 Certification and Completion Proof

Upon successful completion of the 8-week program, weekly assessments, and the Final Assessment Test, I was awarded the Certificate of Virtual Internship for the Siemens Data Science Master track, with an Internship Grade of "O" (Outstanding). The certificate and the corresponding offer letter, both issued in my name, are presented in Section 3 of this report as supporting documentation.

The offer letter also confirms that the program consisted of eight weeks of structured, week-wise learning followed by final credential validation and internship grade point assessment.

5. SKILLS DEVELOPED DURING THE INTERNSHIP

- **Data Science:** Gained a working understanding of the end-to-end Data Science workflow.
- **Data Engineering:** Learned to access, load, transform, and process data for downstream analysis.
- **Machine Learning:** Built both foundational and advanced Machine Learning knowledge.
- **Data Preparation:** Learned data cleaning, transformation, and automation techniques.
- **AI Applications:** Understood how AI is applied to solve real-world business problems.
- **Deployment & Monitoring:** Gained hands-on exposure to deploying and monitoring Machine Learning models.
- **Platform Administration:** Learned to configure and manage an enterprise AI platform (RapidMiner AI Hub).

6. CHALLENGES AND LEARNING EXPERIENCE

One of the early challenges was connecting the different pieces of the Data Science pipeline — data engineering, preparation, modelling, deployment, monitoring, and platform administration — into a single coherent picture rather than treating them as isolated topics. The week-wise structure of the program helped a great deal here, since each module built directly on the previous one. Moving from professional-level to master-level content also required a noticeable step-up in effort, particularly in the advanced Machine Learning and Platform Administration modules, but the weekly assessments kept my learning on track and made sure concepts were reinforced before moving ahead.

7. CONCLUSION

The Siemens Data Science Master Virtual Internship gave me a structured, practical introduction to Data Science and its related technologies. Across eight weeks, it covered Applications & Use Cases, Data Engineering, Machine Learning, Data Preparation, model deployment and monitoring, and Platform Administration — building not just technical knowledge but also analytical thinking, problem-solving, and self-directed learning habits. Completing the program's professional and master-level modules, along with the weekly assessments and Final Assessment Test, resulted in the award of the Certificate of Virtual Internship with an Outstanding grade. Overall, the internship gave me a much clearer picture of how Data Science, AI, and Machine Learning come together in practice.

8. FUTURE SCOPE

The foundation built during this internship opens up several directions for further learning, including Deep Learning, working with larger real-world datasets, cloud-based data engineering, MLOps practices for model deployment and monitoring at scale, and Big Data technologies. I intend to build on this experience to deepen my specialization in Data Science, Machine Learning, and AI-powered applications going forward.

9. BIBLIOGRAPHY / REFERENCES

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